

Data Analytics (DA)

Module-End -3

Module-End Project: Smart Mobility & Ride Analytics

Project Title: Smart Mobility & Ride Analytics

Problem Statement:

Analyze a real-world ride-hailing business dataset using concepts from Mathematics for Data Science, Probability, Statistics, and Hypothesis Testing.

Your goal is to uncover operational insights, validate assumptions, and support decision-making using analytical methods.

Dataset: [Trip Analysis Dataset](#)

Assignment Tasks:

- Prepare and clean the dataset using NumPy and Pandas
- Apply Linear Algebra concepts (vectors, matrices, determinant, eigenvalues/eigenvectors)
- Use Probability concepts (basic probability, conditional probability, Bayes' theorem, distributions)
- Perform Descriptive Statistics (mean, median, mode, variance, standard deviation, skewness, kurtosis)
- Detect outliers using percentiles, quartiles, IQR, and z-score
- Measure relationships using Pearson, Spearman, and covariance
- Apply Inferential Statistics (bootstrapping, CLT, point & interval estimation)
- Conduct Hypothesis Testing (z-test, t-test, chi-square, F-test, ANOVA, A/B testing)

Analysis Questions:

1. Does trip distance significantly affect fare amount?
(Hint: Use **Pearson correlation**, **covariance**, and **linear algebra interpretation**.)
2. Are premium rides statistically more expensive than economy rides?
(Hint: Compare groups using **t-test / ANOVA** and summarize with **descriptive statistics**.)
3. Is there a relationship between surge pricing and customer ratings?
(Hint: Use **Spearman correlation** and test significance statistically.)
4. Do peak-hour rides differ in revenue compared to non-peak rides?
(Hint: Apply **z-test / t-test**.)
5. Are there unusual fare values that indicate anomalies or pricing errors?
(Hint: Detect using **IQR**, **z-score**, **quartiles**, and **percentiles**.)
6. Can ride demand probabilities be estimated for different ride categories?
(Hint: Use **basic probability**, **conditional probability**, and **Bayes' theorem**.)
7. Which operational factors are most strongly related to business performance?
Hint: Use **Pearson/Spearman correlation**, **covariance**, and **eigenvalue analysis**.
8. Can statistical evidence support pricing or service improvements?
Hint: Use **confidence intervals**, **bootstrapping**, and **hypothesis testing conclusions**.

Deliverables

- Python notebook/script with complete analysis
- Clear outputs and explanations
- Final conclusions supported by statistical evidence
- One-page summary of findings and recommendations

Skills to be Evaluated and Mapping

Assignment Section	Core Skill	Sub-Skill	Expected Level
Data Preparation	Python + Pandas	Dataset loading, structure checks, cleaning setup	Beginner
Mathematical Modeling	Linear Algebra	Vectors, matrices, determinant, eigenvalues/eigenvectors	Intermediate
Probability Analysis	Probability	Basic probability, conditional probability, Bayes theorem, distributions	Intermediate
Statistical Analysis	Quantitative Analysis	Descriptive stats, symmetry, dispersion, estimation	Intermediate
Relationship Analysis	Data Analysis	Correlation, covariance, dependency checks	Intermediate
Inferential Statistics	Statistical Testing	Bootstrapping, CLT, confidence intervals, hypothesis testing	Advanced
Reporting & Interpretation	Communication	Conclusions, evidence-based decisions	Beginner

Rubric & Mark Allocation (Total 25 Marks)

Skill / Component	Full Marks	Criteria for Full Marks	Partial Credit Description	Partial Credit
Data Preparation	3	Dataset loaded, reviewed, and prepared correctly	Minor setup or cleaning gaps	1–2
Mathematical Modeling	4	Correct vector/matrix operations with interpretation	Incomplete calculations or weak explanation	2–3
Probability Analysis	4	Accurate probability calculations and distributions	Partial application or missing steps	2–3
Statistical Analysis	5	Correct descriptive stats, outliers, and symmetry measures	Missing some measures or minor errors	3–4
Relationship Analysis	3	Proper correlation and covariance analysis	Limited relationship testing	1–2
Inferential Statistics	4	Valid estimation and hypothesis tests with conclusions	Incomplete tests or weak interpretation	2–3
Reporting & Interpretation	2	Clear insights and actionable conclusions	Basic or unclear summary	1

Skill Taxonomy Structure

- **Data Preparation:** Load, inspect, and clean datasets for analysis
- **Mathematical Modeling:** Apply vectors, matrices, determinants, and eigen concepts
- **Probability:** Solve basic, conditional, and Bayesian probability problems
- **Statistical Analysis:** Measure central tendency, dispersion, symmetry, and outliers
- **Relationship Analysis:** Evaluate correlation and covariance between variables
- **Inferential Statistics:** Perform estimation, bootstrapping, and hypothesis testing
- **Reporting:** Interpret results and provide evidence-based conclusions

Evaluation Guidelines

- Review the complete code, outputs, and explanations.
- Match each section to the corresponding skill and sub-skill.
- Award marks based on rubric standards.
- Use only whole-number partial credits.
- Reduce marks for incomplete logic, missing outputs, or unsupported conclusions.
- Comments must highlight both strengths and areas for improvement.
- Focus on correctness, clarity, and application of concepts—not presentation style.

Achievement Criteria & Passing Rules

- Every section must receive at least partial credit.
- Students must fully achieve at least **two major skill areas**:
 - Mathematical Modeling
 - Probability Analysis
 - Statistical Analysis
 - Inferential Statistics
- Work must be executable, logically correct, and aligned to the case study.
- A detailed scorecard and written feedback are mandatory for every submission.